

2.1

2. (a) For which values n does K_n , the complete graph on n vertices, have an Euler cycle?

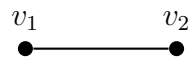
* Euler cycle $\Rightarrow$ complete graph and all vertices of even degree

We know that in any complete graph K_n , the degree of every vertex is $n - 1$. So, if n is odd, $n - 1$ will be even, and every vertex in K_n will have even degree. $\therefore$ any odd integer n will force K_n to have an Euler cycle.

(b) Are there any K_n that have Euler trails, but not Euler cycles?

We know any odd integer n guarantees an Euler cycle. Suppose we choose an even integer n . We know that the degree of every vertex in a K_n will be $n - 1$, which is odd. So, that means in any K_n where n is even and $n > 2$, we will have more than 2 vertices of odd degree. So, the only possible K_n that has an Euler trail but not Euler cycles is

K_2 .

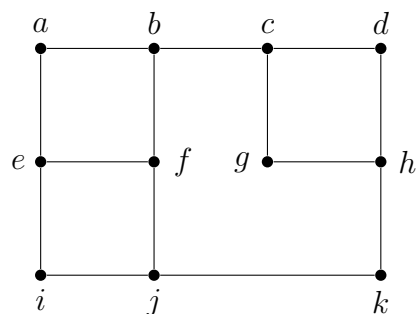


complete graph with exactly 2 vertices of degree 1

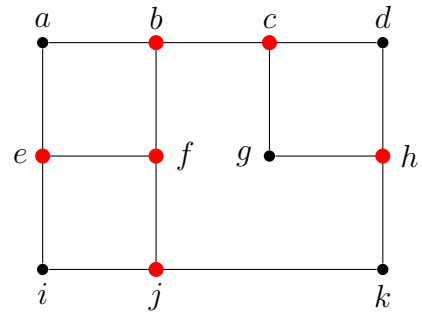
(c) For which values of r and s does the complete bipartite graph $K_{r,s}$ have an Euler cycle?

Since the degree of the vertices on the LHS of $K_{r,s}$ is s and the degree of the vertices on the RHS is r , r and s must be even integers for every vertex in $K_{r,s}$ to have an even degree.

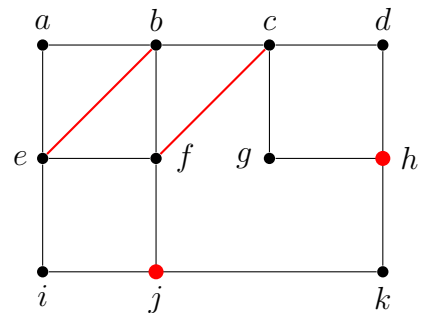
4. What is the minimum number of times one must raise one's pencil in order to draw the graph in Figure 1.4?



To figure out the minimum number of times needed to raise your pencil to draw the above graph, we need to find the minimum number of Euler trails within the graph.



However, we have more than 2 vertices of odd degree: b, c, e, f, h and j . So, adding 2 imaginary edges to the graph,



we now have exactly 2 vertices of odd degree: h and j . Since 2 edges were added for this graph to have an Euler trail, one only needs to pick up their pencil 2 times to draw the graph. The sequence in which this is possible is as follows:

- 1) $c - g - h - d - c - b - a - e - i - j - f - e$
- 2) $b - f - j - k - h$